

EECS 4413 Group Project

Deliverable 3

Purpose of the Deliverable

The purpose of this deliverable is to produce:

- a) The back-end implementation of the use cases, following REST principles and modern runtime deployment requirements.
- b) The front-end and UI of the project
- c) A distinguishable feature for your project
- d) Test cases, delivered as a curl/postman script file

In this Assignment, you will learn to program the full stack of an e-commerce application and to deploy and operate it as containers

Due

Please check e-class regularly for any updates regarding the submission process and deadlines. One submission per group.

Late Penalty

Late project will be accepted for up to two days after the due date. Submissions late up to 24 hours will incur a penalty of 20%. No submissions after 48th hour of the deadline are accepted. The same time windows hold if there is an extension given by the instructor.

Group Effort

This stage of the project is expected to be a group effort, with each member of the group contributing equally. The contribution of each member should be documented. Also, each member should understand the entire project, even if their focus was on specific parts. In the face-to-face demo with the instructor or/and TA, students should be able to explain any parts of the code in their project. **If it is determined that you are guilty of cheating on the assignment, you could receive a grade of zero with a notice of this offence submitted to the Dean of your home faculty for inclusion in your academic record.**

What to Hand in

Your submission, as noted above, will be electronically in the form of an archive (e.g. zip) file. Please check e-class regularly for any changes related to the submission process. You are to complete the deliverable by providing.

-the server and client-side implementations as a zip file (Maven project);

-a readme file with installation instructions; if you host your application at an external site, please provide the url as well as any credentials needed to operate it

-A script file with curl or postman test cases for the backend; the testcases should test for robustness (user wrong inputs), scalability and security.

-an updated design document that explains the distinguishable feature, changes in the design, testcases, project plan and team meeting logs as well as the performance and security/testing report. Append the new content to Deliverable 2 document.

Note: you are asked to provide a distinguishable feature for your project. This is in addition to original project specifications. The use case can be

- A Gen AI use case (see the lecture and tutorial on AI and MPC)
- . For example, you might want to create a chat bot that, either
 - Allow users to use natural language to interact with your application
 - Allow a business admin to chat with the system about the business aspects of the e-auction site (bid statistics)
 - Allow the DevOps team to monitor the application (logs and metrics), assists in mitigation strategies
- A feature that improves the quality of your application. For example, you can have logging, instrumentation for observability and you can make your application self-adaptive for performance or availability using AI or non-AI techniques.
- Any other feature that you might choose. Discuss it with the instructor and TAs

Development and Runtime Frameworks

For Deliveries 3, you are allowed to use any technology presented in class and encouraged to go beyond the APIs covered in Labs and lectures. You can also use other technologies, not presented in the class. For example, you can use React (<https://react.dev/learn>). However, it is expected that each member of the team can contribute to and understand each part of the project and therefore agrees on the technologies used in the project. If you use other technologies, you should discuss it with the TA or the professor.

Team Development and Deployment

Ideally, the application has multiple microservices accessed through a web gateway.

- To be considered microservices based, the application should, at minimum, have the middle-tier microservice based, with REST APIs, testable with curl/postman scripts.
- It is required that you show your final microservice architecture in the design document.

Ideally, the application is packaged in Docker containers, can be pulled from Docker repository, and run.

- To be considered cloud enabled, at minimum, the backend is packaged into a container that can be pulled and run using docker commands.
- It is highly recommended that you show your deployment diagram in the design document

The front end, UI, must be functional and allows easy navigation between different roles and use cases. Although the style, look and feel, are very important in modern e-commerce applications, they are not the focus of this course.

If you use GitHub, provide the link to it and give your TA access to it. Otherwise, provide your source files as a zip file. The TA should see your code and scripts.

In Person Demo

For in person demo, you will meet the TA or professor. Be ready to show:

- how you deploy and run your application
- highlights the use cases for the application
- explain the design document and its sections
- answer any questions about the code, architecture, deployment